1. Problem and Motivation

We study gender bias in static word embeddings (GloVe 6B 300d). Most prior work assumes that gender bias can be represented as a single direction in embedding space and removed by projection.

In this work, we investigate whether gender bias is truly one-dimensional or whether it is distributed across multiple principal components of a gender subspace.

Our approach is to progressively remove principal components derived from gender-definitional word pairs and observe how different bias and geometry metrics respond.

2. Experimental Setup

We construct a gender subspace using PCA on difference vectors of definitional word pairs (e.g., he–she, man–woman, king–queen).

We then progressively remove the top k principal components from all word vecto

$$(\boldsymbol{v})^{\boldsymbol{k}} = (\boldsymbol{I} - \sum_{\boldsymbol{i}=\mathbf{1}}^{\boldsymbol{k}} \boldsymbol{g}_i \boldsymbol{g}_i^{\boldsymbol{T}})\boldsymbol{v}$$

We evaluate the following metrics:

- Direct Bias (projection-based metric)
- WEAT association score (Cohen's d)
- Mean vector displacement
- Neighbor stability@10

We test k ∈ {0, 1, 2, 3, 5, 8, 10}.

3. Main Observations

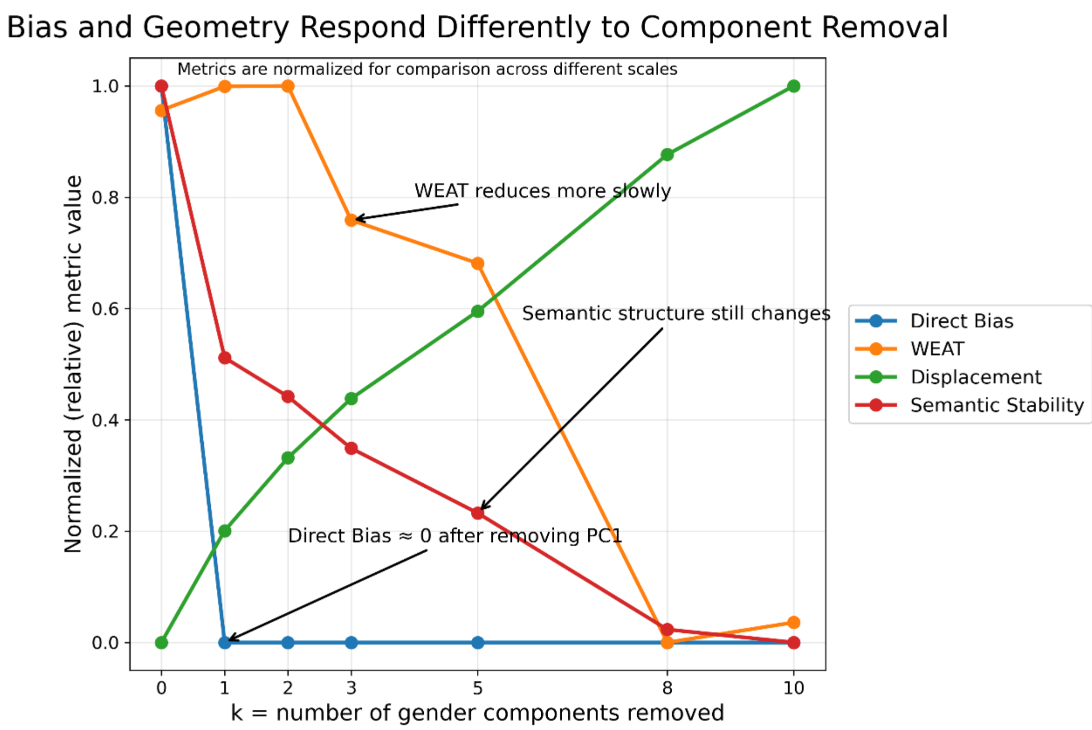

Figure 1. *Progressive removal of gender principal components.*

Key observations:

- Removing PC1 eliminates Direct Bias almost completely.
- WEAT decreases more gradually.
- Semantic geometry (displacement and neighbor stability) continues to change as more components are removed.

4. Mathematical Formulation of Debiasing Operator

We formalize the removal of principal components as a linear projection operation.

Let $\boldsymbol{v}\ \boldsymbol{in}\ \boldsymbol{R}^{\boldsymbol{d}}$ be a word embedding vector and let $\left\{\ \{\boldsymbol{g}_i\}_{\{\boldsymbol{i}=\boldsymbol{1}\}}^{\{\boldsymbol{k}\}}\right\}$ be orthonormal principal components derived from the gender subspace. Removing a single principal component g_i is defined as:

$$\boldsymbol{v}^{\{(i)\}} = \left(\boldsymbol{I} - \boldsymbol{g}_i \boldsymbol{g}_i^T\right)\boldsymbol{v}$$

More generally, removing the first k components:

$$(\boldsymbol{v})^{\boldsymbol{k}} = (\boldsymbol{I} - \sum_{\boldsymbol{i}=\boldsymbol{1}}^{\boldsymbol{k}} \boldsymbol{g}_{\boldsymbol{i}} \boldsymbol{g}_{\boldsymbol{i}}^{\boldsymbol{T}})\boldsymbol{v}$$

This operator corresponds to projecting vectors onto the orthogonal complement of the subspace spanned by the selected principal components.

5. Single Principal Component Ablation

To better understand how bias is distributed across principal components, we perform a *single-component ablation study*. Instead of removing the top $k$components cumulatively, we remove each component individually:

$$\boldsymbol{v}^{\{(\boldsymbol{i})\}} = \left(\boldsymbol{I} - \boldsymbol{g}_{\boldsymbol{i}} \boldsymbol{g}_{\boldsymbol{i}}^{\boldsymbol{T}}\right)\boldsymbol{v}$$

For **i = 1, …. 10** for each modified embedding, we evaluate:

- Direct Bias
- WEAT score
- Mean displacement from the original embedding
- Neighbor stability@10

6. Key Results from Principal Component Ablation:

GitHub: https://github.com/AlexeyKresin/embedding-bias-geometry/tree/main/experiments/single%20pc%20ablation

Our results reveal a strong asymmetry between bias removal and geometric distortion:

- **PC1 dominates Direct Bias**
  Removing the first principal component reduces Direct Bias to nearly zero.
- **WEAT is not aligned with PC1**
  The largest reduction in WEAT occurs at higher components (e.g., PC4), indicating that bias is distributed across multiple directions.

- **PC1 causes maximal geometric distortion**
  Removing PC1 results in the largest mean displacement, suggesting it encodes substantial non-bias-related information.
- **Higher components produce smaller, more stable changes**
  Removing PCs beyond PC1 leads to moderate geometric changes while still affecting bias metrics.

7. Interpretation

These results suggest that gender bias in word embeddings is not strictly one-dimensional.

While Direct Bias is concentrated in the first principal component, association-based metrics such as WEAT depend on multiple directions in the embedding space.

This indicates a trade-off:

- Removing dominant components (e.g., PC1) effectively reduces bias but introduces significant geometric distortion.
- Removing smaller components leads to less distortion but only partial bias reduction.

This highlights a fundamental tension between fairness and semantic preservation in embedding debiasing. These findings challenge the common assumption that bias can be removed by eliminating a single direction, and instead suggest a more distributed structure of bias in embedding space.